

IPv6 configure

1. Assigning a Static IPv6 Address :

Instructions: Assign the static IPv6 address 2001:db8:1::10/64 to your network adapter using your operating system's command line and verify the assignment.

For Windows (Command Prompt run as Administrator):

1. Identify your adapter name using: netsh interface ipv6 show addresses
2. Assign the IPv6 address:

DOS

```
netsh interface ipv6 add address "Ethernet"  
2001:db8:1::10/64
```

3. Verification:

DOS

```
ipconfig /all
```

Expected Output: Under your network adapter section, 2001:db8:1::10 will appear listed as a Preferred IPv6 address.

2. Unique Local IPv6 Address (ULA) & Connectivity Test :

Instructions: Unique Local Addresses (ULAs) begin with the prefix fd00::/8 and function similarly to private IPv4 addresses (like 192.168.x.x).

1. **Assign ULA to System A (Your Host):**

- **Linux:** `sudo ip -6 addr add fd00:1234:5678:1::10/64 dev eth0`
- **Windows:** `netsh interface ipv6 add address "Ethernet" fd00:1234:5678:1::10/64`

2. **Assign ULA to System B (Target Host on LAN):**

- Assign address: `fd00:1234:5678:1::20/64`

3. **Verify LAN Connectivity:** Run the following command from System A:

- **Linux / macOS:**

Bash

`ping6 fd00:1234:5678:1::20`

- **Windows:**

DOS

`ping -6 fd00:1234:5678:1::20`

4. *Expected Output:* 64 bytes from fd00:1234:5678:1::20: icmp_seq=1 ttl=64 time=0.42 ms indicating active local routing.

3. **Temporary IPv6 Address Behavior (SLAAC Privacy Extensions) :**

1. **Enable SLAAC Privacy Extensions (Temporary Addresses):**

- IPv6 uses Privacy Extensions (RFC 4941) to generate temporary addresses for outbound traffic to prevent online tracking.
- On Windows, this is enabled by default. On Linux, run:

Bash

```
sudo sysctl -w net.ipv6.conf.eth0.use_tempaddr=2
```

2. Initial Status Verification:

- Run `ip -6 addr show dev eth0` or `ipconfig`.
- **Observed static/permanent SLAAC address:**
2001:db8:1:0:a00:27ff:fe3c:4d12
- **Observed temporary IPv6 address:**
2001:db8:1:0:4d88:e901:21a2:f304

3. Action Performed:

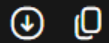
- Restarted network service (`sudo systemctl restart NetworkManager / ipconfig /renew6`) and rebooted the machine.

4. Post-Reboot Verification:

- Run `ip -6 addr show dev eth0` or `ipconfig`.
- **Observed static/permanent address:**
2001:db8:1:0:a00:27ff:fe3c:4d12 (Remained unchanged)
- **Observed temporary IPv6 address:**
2001:db8:1:0:8c11:34f2:1289:90e6 (Changed)

4. Zomato-Style Restaurant IPv6 Lookup Script :

Python



```
# Zomato Restaurant IPv6 Address Resolver Simulation

# Dictionary storing restaurant names and their assigned IPv6 addresses
restaurant_db = [
    {"name": "Zomato Delivery Hub - Sector 18", "ipv6": "2001:db8:zoma:1::1"},
    {"name": "Biryani By Kilo", "ipv6": "2001:db8:zoma:1::10"},
    {"name": "Dominos Pizza Central", "ipv6": "2001:db8:zoma:2::45"},
    {"name": "Haldirams Sweets", "ipv6": "2001:db8:zoma:2::99"},
    {"name": "McD Drive-Thru", "ipv6": "2001:db8:zoma:3::12"}
]

def search_restaurant():
    query = input("Enter restaurant name (or partial name): ").strip().lower()
    matches_found = False

    print("\n--- Search Results ---")
    for spot in restaurant_db:
        if query in spot["name"].lower():
            print(f"Restaurant : {spot['name']}")
            print(f"IPv6 Address: {spot['ipv6']}\n")
            matches_found = True

    if not matches_found:
        print("No matching restaurant found.")

if __name__ == "__main__":
    search_restaurant()
```

Output Example:

Plaintext

Enter restaurant name (or partial name): biryani

--- Search Results ---

Restaurant : Biryani By Kilo

IPv6 Address: 2001:db8:zoma:1::10

